PART 4 | Trauma

CHAPTER

26 Introduction to trauma

Learning objectives

- Become familiar with the timeline concept in trauma management
- Understand how to assess a trauma problem
- Appreciate the importance of observing trends for identification of evolving conditions
- Recognise that specific criteria exist for implementation of different treatment strategies

DEFINITION OF TRAUMA

Trauma can be defined as an injury to any part of the human body as the result of energy transfer from an inflicting source. The forces that can lead to injury include chemical, thermal, ionising radiation and mechanical. The extent and severity of the trauma sustained depends upon the magnitude, nature and duration of the inflicting cause.

Major trauma denotes injuries to more than one body region or organ system. In Part 4, trauma will be examined from a variety of viewpoints, interconnected to different specialties. In this chapter we will examine the facets that bind the whole topic together.

THE MAGNITUDE OF THE PROBLEM

In western industrialised countries, trauma accounts for the largest number of deaths and disability in children and young adults. According to the World Health Organization, there are an estimated 5 million injury fatalities worldwide, representing about 9% of global deaths. This rate is 1.7 times higher than deaths caused by malaria, tuberculosis and human immunodeficiency virus (HIV)/acquired immunodeficiency syndrome (AIDS).

Road traffic accidents (RTAs), falls and intentional violence continue to be the most prevalent causes of trauma fatalities, with a combined rate of 64%. Of note, the major burden of injury is increasing in low- and middle-income countries secondary to industrialisation and increasing motorised transportation. Interestingly, the annual incidence and trends over time show that there is variation from country to country. Information is usually obtained from national statistics organisations that use the International Classification of Diseases, a system that can be limited in providing accurate descriptions of injury severity. In contrast, the Abbreviated Injury Scale dictionary consists of a greater level of detail (including more than 2000 injury codes) and assigns to every injury a severity score between 1 (mild) and 6 (maximum). This can be summated into the so-called injury severity score (ISS), which provides an indication of the anatomical severity of injury suffered by the individual patient. Major trauma is defined as an ISS greater than 15. The majority of patients admitted to hospital with injury have low ISS values, ranging between 4 and 9, with injuries such as an isolated limb fracture and/or isolated mild head injury. Overall, major trauma affects approximately 15% of all injured patients.

According to the UK Department of Transport, in 2019, the overall number of casualties following an RTA reported to the police, including all severities, was 153 158, of whom 25 945 had sustained severe injuries. Overall, 1752 individuals died as a result of their injuries. Across Europe, according to the data presented by the European Transport Safety Council's Performance Index (PIN) report, it appears that the overall number of fatalities was reduced by 3% compared with 2018, and was estimated to be 22 660. In 16 countries, the death rate decreased whereas in 12 it increased, with the four remaining countries registering no change.

As countries do not use the same definition of serious injury, international comparisons are based on road deaths per million inhabitants. In the 27 Member States of the European Union (EU27), the overall level of road mortality was 42 deaths per million inhabitants in 2020 compared with 67 per million in 2010 (***Figure 26.1***). The EU road mortality rate was 51 per million in 2019; the unprecedented drop in mortality between 2019 and 2020 was mainly due to the traffic restrictions to contain the pandemic.

Norway remains the leader among PIN countries with 17 road deaths per million inhabitants, followed by Sweden with 20 deaths per million inhabitants in 2020. In Malta, the UK, Switzerland and Denmark, road mortality is below 27 per million. The highest road mortality is in Romania and Latvia, with 85 and 73 road deaths per million inhabitants, respectively.

A large proportion of severely injured survivors experience long-term or permanent disability as a result of their injuries. Approximately one-third of severely injured survivors sustained life-changing injuries and were unable to return to

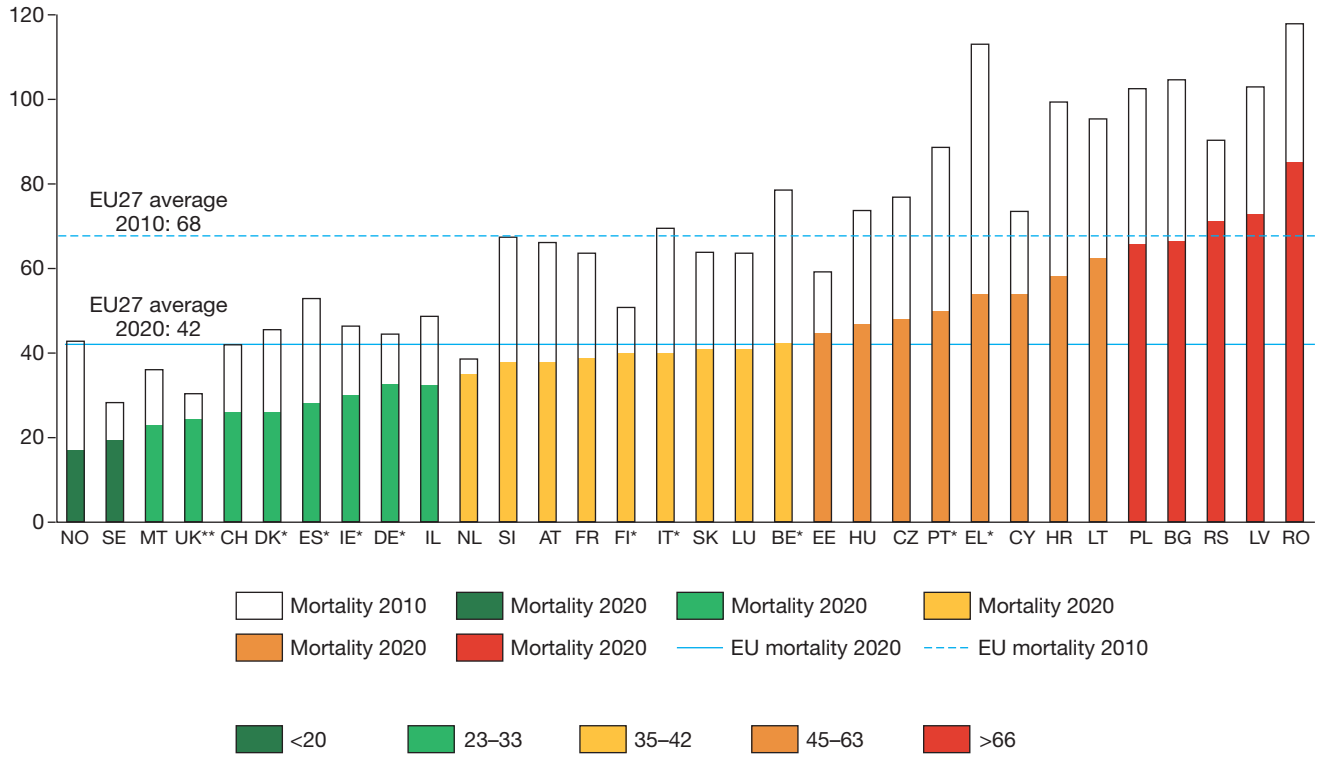

Figure 26.1 Mortality (road deaths per million inhabitants) in 2020 (coloured bars) with mortality in 2010 for comparison (white bars) (courtesy of the European Transport Council). *National provisional estimates used for 2020. The annual number of deaths in LU and MT are particularly small and, therefore, subject to substantial annual fluctuations. **UK data for 2020 are the provisional total for Great Britain for the year ending June 2020 combined with the total for Northern Ireland for the calendar year 2020.

their previous level of function and occupation, undergoing profound lifestyle changes including long-term pain and suffering. It should be emphasised that an injury affects not only the injured person but also everyone who is involved in the injured person's life. The impact of the modern epidemic of RTAs on the universal epidemic of violent injury cannot be overstated. The annual direct medical cost of injuries treated in hospitals and additional care facilities is estimated to be somewhere between £3.5 billion and £4 billion. Moreover, additional indirect costs, due to loss of earnings, loss of productivity and a reduced quality of life, increase the total sum significantly.

Multiple injuries are often thought to occur primarily in younger patients who are involved in incidents with a high energy transfer. Lately, multiple injuries are also seen in the older population as patients live longer, are more active and we have better diagnostics. The last point is particularly important as older adult patients were often underinvestigated, so the full extent of injury was not appreciated.

Recent data published by the UK's National Institute for Health and Care Excellence (NICE) revealed that there are an estimated 500 000 new patients with fragility fractures per year in the UK; approximately 70 000 of these are admitted with proximal femoral fractures, among whom 6.7% of those aged >65 years will die within 30 days of the incident, as reported by the National Hip Fracture Database. Most of the remaining patients with proximal femoral fractures will have diminished independence and functional capacity. It is therefore unsurprising that this particular cohort of patients, the number of whom will increase in coming years owing to the anticipated increase in life expectancy, is thought to represent a huge burden on healthcare services and society in general.

In summary, the overwhelming majority of trauma is not life- or limb-threatening and full recovery with return to preinjury status is usually expected. However, it remains of paramount importance that injuries are detected early in order to properly intervene and therefore achieve a favourable outcome. For instance, one must be vigilant not to miss paediatric non-accidental injury (NAI) (see ***Chapter 44***) or injuries secondary to a chronic underlying disease process rather than the injury itself, for example pathological fractures in older age groups. Of note, it has been shown that, in 66% of cases when children die as a result of abuse, there has been some previous interaction with a health professional or with social services, but the seriousness of the situation was not fully appreciated.

Summary box 26.1

Trauma: the magnitude of the problem

- The vast majority of trauma is not life- or limb-threatening
- Severe trauma continues to be a major cause of death in young patients
- Older adult patients with fragility fractures are a special group posing a further burden to a healthcare system
- Promptly identifying important features of injuries could influence the outcome

THE MANAGEMENT OF TRAUMA

As soon as a severe injury occurs, every second counts, and all aspects of decision making and management are critical for a patient's long-term quality of life and even their survival. The concepts of initial assessment and management have specific goals that are based on practice over a long period of time. In the modern era, new protocols have been formulated that are centred on a profound understanding of the physiology of the host response to an acute threat to homeostasis; these protocols allow clinicians to use standardised measures and to speak a common language. They also reduce delays and expedite patient care, especially when clinicians are under pressure to make a critical decision. However, an understanding of the reasoning behind them remains crucial.

As in other acute conditions, the patient is particularly reliant upon the clinician when trauma occurs. A patient with a chronic condition is familiar with the nature of their problem and the way in which it is progressing. The surgeon may offer a remedy and the patient may consider the potential benefits and choose appropriately whether to accept the remedy. The injured patient does not know what will happen without treatment and so relies on the surgeon to inform them of both the natural history and the potential benefits of any intervention. The implication is that, as surgeons, we have a duty to be aware of both.

The significance of time in the outcome

Injuries can happen at lightning speed. Time point 0 (time 0) is defined as literally the seconds prior to the event, when the patient is at their normal baseline. All subsequent events, including the acute physiological response to injury, the body's internal mechanisms to maintain homeostasis (to compensate for the sequelae of trauma), the healing processes and the actions instigated by health professionals, are associated with a 'timeline'. This 'timeline principle' is crucial to a deeper understanding of how to prioritise assessment, investigation and treatment in what may be a rapidly evolving situation following injury. There is an optimal time window during which an intervention can have a radically positive effect on treatment outcome. Based on this timeline, interventions may be grossly categorised as emergent (life-saving), acute (restoring and maintaining physiological and physical stability) and delayed or semielective (focusing on the treatment of post-fracture fixation complications [non-union, infection and malunion from the orthopaedic trauma point of view]).

In the immediate aftermath of a major trauma, the physiological crisis continues to evolve, the risk of death is increased and less appropriate and prompt interventions are carried out. Potentially rapidly evolving situations, such as airway obstruction, tension haemothorax and haemopericardium, if left untreated, will inevitably have catastrophic consequences and therefore should be given priority in terms of the initial medical response to an injured patient. Thus, the seriousness and the immediate impact of a specific clinical condition should dictate its prioritisation, leading to a systematic approach ('what kills first should be managed first') (***Figure 26.2***).

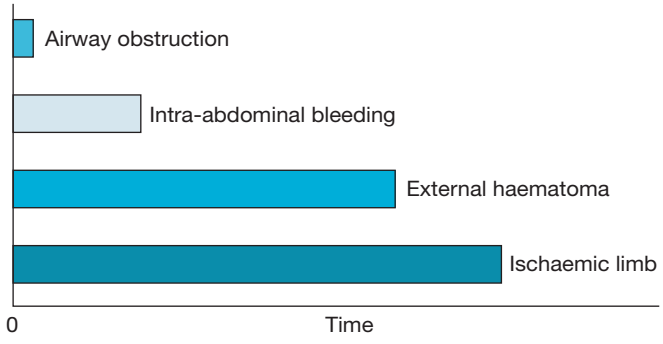

Figure 26.2 Estimated time from incident to death or irretrievable damage for various conditions.

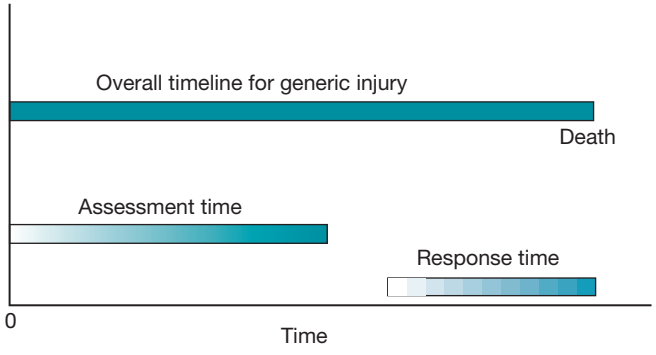

Figure 26.3 Diagrammatic representation of the relationship between assessment and response times. In this example, there is time to assess and respond effectively before death.

The Advanced Trauma Life Support (ATLS) system delineates an order of priorities defined by ABCD; that is, airway, breathing, circulation and disability (neurology). This hierarchy of priorities is based on the 'time dependence' principle. In other words, the time taken to manage an individual problem is the sum of the time taken to identify it and to execute effective treatment (***Figure 26.3***). In such settings, time is critical, so the normal history and physical investigations are not performed during the ATLS primary survey, but the primary focus is on detecting and identifying individual problems, ranking them in order of priority and dealing with them effectively and efficiently in their appropriate timeframes.

The clinician should take into consideration the mechanism of injury and initial clinical findings then promptly request and carry out specific investigations, for example computed tomography (CT) scans. This is to allow rapid and precise identification of injuries that may benefit from early therapeutic intervention and that otherwise might be clinically challenging as the initial signs may be subtle or non-specific. This proactive approach is critical, as the evaluation and diagnosis of an important injury may be difficult before the full-blown and potentially life-threatening presentation of that injury.

A typical example would be an RTA victim with a scalp laceration and a reduced Glasgow Coma Scale (GCS) score of 13/15; such a drop in the GCS could be explained by head

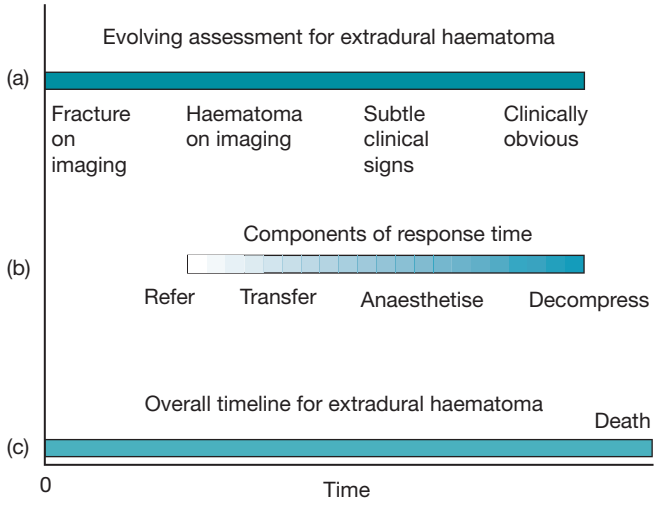

Figure 26.4 Diagrammatic representation of the relationship between assessment and response times for extradural haematoma: **(a)** the stages of assessment, **(b)** the components of the response and **(c)** the overall time from incident to death. It can be seen that relying on obvious clinical signs gives insufficient time to respond effectively.

injury or by the presence of shock and hypoxia. Waiting for the development of overt clinical signs could be dangerous. Should there be any injury such as an extradural haematoma this can be diagnosed with a head CT prior to the presentation of clinical signs. Prompt surgical decompression will result in a reduced risk of morbidity and mortality. In this situation, if the time taken to make the diagnosis were prolonged, such that the clinical signs presented prior to treatment intervention, it may be too late to prevent the death of the patient (***Figure 26.4***). This frequently encountered clinical case scenario demonstrates the principle that we need to introduce a management response even before we have made the definitive diagnosis if we want to save the patient's life. It is clear, therefore, that the 'timeline concept' is critical in the safe management of trauma patients.

Reducing the diagnosis time and response time of our interventions is dependent not only on clinical staff but also on the availability of resources. For the example above, a dedicated trauma team and a CT scanner should be available 24/7. Recently, this 24/7 availability of the trauma team and the designation of regional hospitals to operate as Level I Trauma Centres, with the availability of all disciplines and appropriate equipment on site, has provided the necessary foundation for the development of a unified trauma care system in England. Indeed, the first reports published on its effectiveness in saving lives have been very positive.

The 'timeline concept' that has been discussed in the management of patients with multiple trauma can be applied to patients with isolated injuries. Again, the key issue – irrespective of the number of injuries – is to minimise delays in making the diagnosis and promptly initiate treatment. Such a global approach would save lives, reduce morbidity and make the healthcare system more efficient in terms of resource utilisation, as well as cost-effectiveness.

Summary box 26.2

The importance of time

- A thorough understanding of the 'timeline concept' in trauma is critical
- Assessment should be completed within a set time
- The time to respond is limited
- The goal is for both assessment and response to take place in the time window prior to irreversible damage or death

Most importantly, dealing with a patient in an acute setting is a dynamic situation, which may fluctuate unpredictably along the timeline. Therefore, any observation and analysis may evolve rapidly to an extent that interventions need to be modified accordingly. Thus, ongoing patient evaluation is essential in order to identify and respond to every fluctuation noted in a timely fashion. The initial primary survey, applied according to the ATLS protocol in trauma patients, should be followed by secondary and tertiary clinical assessment, even after the acute phase of treatment has been completed successfully. Ongoing monitoring of vital organ activity, ordering of the necessary biochemical and radiological investigations and recording of all the findings in a single place can allow easier evaluation and identification of trends over time to facilitate prompt intervention. Such a strategy may reduce the risk of having undiagnosed injuries and delays in a patient's treatment. Several studies have been published that report on missed injuries and make some recommendations on how to avoid these.

The timeline following an injury is continuous, and the accumulated documentation may become voluminous, complex and confusing. It is helpful periodically to make the effort to stand back and summarise the situation. In its recent trauma guidelines, NICE refers to this and advises that a plain language summary of the situation directed at the patient's family doctor, but intelligible and available to the patient or carers, should be produced within 24 hours.

ASSESSMENT AND RESPONSE

The assessment of trauma

Traditionally, and especially when learning the theory of the various component parts of the assessment of the injured patient, this is done in a sequential linear fashion. However, when dealing with a severely injured patient in practice, time is of the essence and when resources allow several things may be happening simultaneously – this is so-called horizontal management; while there may be a number of practical activities occurring simultaneously, the coordination, assessment and control of the situation remains as a mental exercise for the 'hands-off' team leader. In particular, acquiring very early definitive diagnoses with CT is increasingly important. To provide such a level of care generally needs a systematic approach to concentrating resources and to preferably bringing the severely injured patient to that resource – a major trauma system.

The objective of assessment of a trauma patient is to form the best understanding possible of the injuries sustained and their consequences. Forming that whole picture is a challenging process. The clinician must make the best use of what they know of the mechanism of injury, pre-existing patient factors and the injuries already found. Synthesising the associations of these three factors will help direct attention to finding other less obvious problems. Sometimes, the pieces of the puzzle do not 'add up'; at that point, the clinician should attempt to think 'outside of the box' as there may be something unexpected, such as an unknown underlying pathology leading to a pathological fracture, or deliberately concealed, such as in NAI.

For instance, a 50-year-old male restrained passenger in a car involved in a head-on collision with another vehicle may sustain rib fractures, a sternal fracture and a thoracic spine fracture but there may also be cardiac contusion and abdominal injuries. Very early CT and echocardiography may decrease the need for clinical acumen, but in many centres it is not available. The clinician then needs to use their knowledge of the mechanism of injury and the obvious rib and sternal fractures to direct their attention to excluding the less obvious but potentially life-threatening abdominal or cardiac injuries. Similarly, when sophisticated investigations and techniques are available, the clinician should not be over-reliant on them and neglect the important logical thought processes required to piece together the information that is available. It remains of value to consider how a clinician can make best use of the available information for the benefit of the patient.

Summary box 26.3

The assessment of trauma

- Appreciate the factors in the relationship: mechanism + patient factors = injuries sustained
- Use this to allow obvious features to lead to the discovery of less obvious injuries
- When the features do not appear to 'add up' this should be an alert for the clinician to think 'outside the box' and connect the dots

Mechanisms

Mechanisms may be broadly classified as blunt, penetrating or even of a combined nature. *Table 26.1* shows examples of how knowledge of the mechanism may help detect more covert injuries. Early definitive imaging may be thought to reduce the need for such considerations, but the clinician can help the radiologist by directing their attention to potential diagnoses. The adage that 'unless you are very fortunate you only find what you look for' is apt. The most common mechanism involves blunt trauma, which may be either direct or indirect. In a direct mechanism, the damage is localised to the initial site of that mechanism. In contrast, in an indirect mechanism the damage occurs at a distant site after transmission of the force exerted. For example, a direct kick to the medial aspect of the mid-shaft of the tibia in a footballer by an opponent may induce an isolated transverse tibial fracture. There will be localised bruising and ecchymosis where the force was applied.

Figure 26.5 The injury force in a car accident can be transmitted from the dashboard to the knee and then to the hip, which is the site of injury.

On the other hand, a fall from a height of 1 metre with a twisting moment as the foot hits the ground can lead to a spiral fracture of the distal tibia. In this situation the force is transmitted through the body's tissues to a location some distance away from its original point of application. In this case, other injuries may have occurred along that line of transmission and should also be sought, such as a remote fibular fracture. Similarly, a motor vehicle crash associated with a point of impact of the knee joint of the driver on the dashboard of the car could induce a fracture dislocation of the acetabulum and hip joint (transmission of force from the knee joint to the hip socket – an indirect blunt mechanism) (*Figure 26.5*).

TABLE 26.1 Examples of patterns of injury.

Mechanism	Obvious features	Covert injuries
Left-sided impact from road traffic accident	• Lateral compression of the pelvis • Left-sided pneumothorax	• Splenic rupture • Extradural haematoma
Flexion distraction (lap belt)	• Chance fracture of the lumbar spine • Dislocated knee • Head injury	• Duodenal rupture • Popliteal artery disruption • Cervical spine fracture
Electrocution	Burn on hand and collapse	Posterior dislocation of the shoulder
Dashboard impact	Knee wound	Posterior dislocation of the hip

The 'timeline concept' previously discussed should not be taken to imply that every injury needs urgent treatment or that every injury of a similar type is of equal urgency. For instance, a tibial fracture that a footballer has sustained may be treated satisfactorily for some time after the moment of injury but, in

contrast, a fracture of the acetabulum with a hip dislocation (as described above) represents an emergency owing to the potential development of neurovascular complications (damage to the sciatic nerve; avascular necrosis of the femoral head). Therefore, the clinician's decision-making process should be informed by the peculiarities of the type of injury sustained and the anatomical location involved.

Moreover, it should be appreciated that the conduction of energy in an indirect mechanism, which is transferred via the soft tissues or fluid, can be difficult both to understand and to diagnose (accurately and promptly). For example, the rise in pressure secondary to a lower abdominal force could be passed to the vascular tree (aorta), leading to unexpected haemorrhage and death. Therefore, one can argue that the effects of direct mechanisms are easier to comprehend than those of indirect ones.

Penetrating injuries are caused either by sharp objects or by firearms (see ***Chapter 34***). When dealing with sharp objects, it is necessary to take into account their length, surface area and the size of the entry point. For example, a pair of scissors will cause damage to the underlying tissues that they contact (skin, subcutaneous fat, fascia, etc.). Local examination will confirm the extent of the injury and the need for wound exploration. It is critical to be familiar with the relevant anatomy of the area involved to accurately assess peripheral nerve function and tendon and muscle integrity. Here again the 'timeline concept' of prompt assessment and response (treatment) can be crucial in cases where there is vascular injury, a compartment syndrome due to internal bleeding or even joint penetration that could lead to septic arthritis. Knowledge of the anatomical structures at risk is essential to making the right decision in a timely fashion. This is particularly critical for penetrating wounds over the torso (see ***Chapter 29***) because it is not always easy to establish the track that the sharp object has followed. In this context, it should not be forgotten that the abdominal structures extend higher than anticipated, and as high as the level of the fifth rib in expiration.

Summary box 26.4

Sharp object injuries

- Length of the sharp object involved is important
- Familiarity with local anatomy is essential
- Crucial point: abdominal structures at risk of injury extend high into the chest

Penetrating injuries caused by firearms are more difficult to understand than incisional injuries caused by sharp objects. A low-velocity projectile may cause similar injuries to a knife, whereas a high-velocity projectile (bullet) causes extensive damage to the tissues as it travels, inducing lateral acceleration far from the point of impact and producing either a permanent or a temporary cavity (see ***Chapter 34***). The importance of the temporary cavity is that it lasts for only milliseconds and is usually not evident during the clinical examination. It is important to be aware that this temporary cavity usually extends far from the boundaries of the apparent injury and there may be ingress of foreign material. Awareness of this phenomenon will ensure that the surgeon carries out sufficient exploration and wound excision.

Summary box 26.5

Firearm injuries

- Passage of high-velocity bullets induces permanent and temporary cavitation
- Temporary cavitation can contain foreign material, but disappears after a few seconds and is less evident to the clinician
- Low-velocity bullets induce similar damage to knives

Patient factors

Every patient is different: each possesses a unique profile and medical history and so will respond differently to a given traumatic incident. Of note, age plays an important role in this regard. Children and adults of different ages will sustain different injuries as a result of the same mechanism. For instance, a car hitting a pedestrian will induce different injuries in an adult from those in a child (***Figure 26.6***). It is important to consider the other aspects of the patient's history: past medical history, medication and allergy risk will direct not only the clinical assessment but also the treatment.

Obvious injuries

Some injuries are very obvious and can be identified before details of the mechanism or patient are known. One can take advantage of this, as the presence of an obvious injury can inform and lead to the identification of another that is less obvious. Obvious injuries are usually visible externally. It is therefore unsurprising that at the end of the ABCD protocol there is also an E, referring to exposure and the need to look for other signs of injury. Bruising to the scrotum of a motorcyclist following a collision with a car suggests a pelvic fracture. Contusion over the greater trochanter of the proximal femur in an older patient experiencing difficulty with straight leg raise points to a fracture of the neck of the femur. Finger-shaped bruises on a child's arms or thighs suggest NAI. The presence of a seat belt mark on the lower abdomen of a patient involved in a car crash and who has substantial abdominal pain points to damage inside the abdomen. Thus, exposure of the trauma patient should be routine practice in order to avoid missing the 'obvious'.

Hidden factors

Mechanisms

Sometimes, the formula 'mechanism + patient = injury' does not seem to 'add up'. If this is the case, the clinician should look further as hidden information may be contained in the mechanism. Occasionally, there is a deliberate attempt to misinform. While the majority of alert and orientated patients tell the truth, other patients, in order to protect themselves or others, may fabricate a mechanism. This may mislead the clinician and guide them to look for the wrong pattern of injuries. For instance, a patient in their twenties with a calcaneal fracture

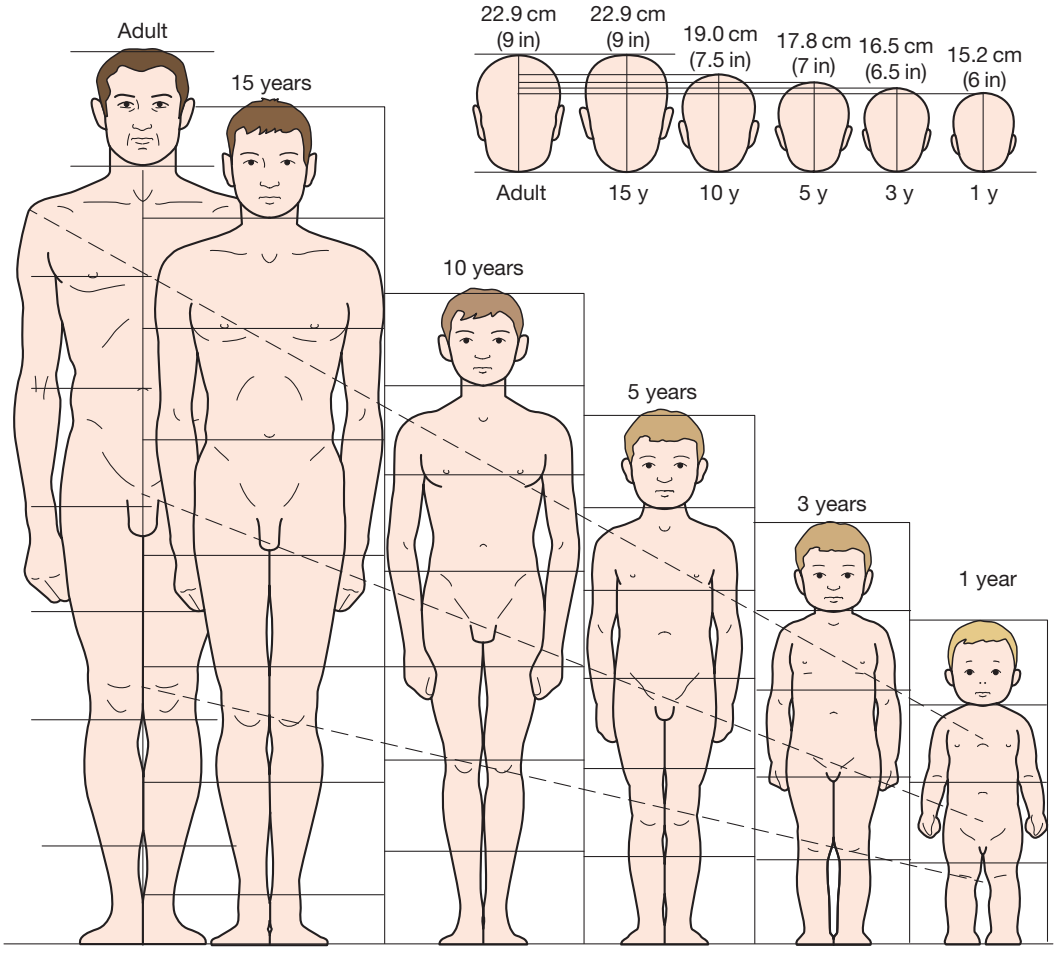

Figure 26.6 Body proportions at various ages and the anatomical location of injuries when hit by a car.

may report that this was the result of a fall into a pothole in the road, when in fact it had occurred during a burglary following a fall from a height of 10 metres. This can delay the accurate diagnosis of the specific injury and may prevent the diagnosis of other important injuries, such as a lumbar spine fracture. Although the patient should be given the chance to tell their story, it should not always be believed, particularly if there are inconsistencies.

A similar situation may arise when a patient is unable to give their history of events, for instance patients who are unconscious. In this scenario, the mechanism of injury is missing. The physically and mentally vulnerable include older patients, perhaps with dementia, and very young children. The difficulty or inability to report the injury is compounded by the fact that it might relate to criminal activity (e.g. NAI). Parameters that should alert the clinician and raise suspicion of NAI include:

- external signs of injuries not consistent with the mechanism reported;
- long bone fractures in a preambulatory child;
- inconsistent or changing history;
- aggressive or unusual behaviour of carers at interview;
- posterior rib injuries.

Summary box 26.6

Hidden mechanisms

- The vast majority of conscious patients will tell the truth
- Patients involved in criminal activity may not tell the truth
- Fear of abuse may prevent vulnerable patients from telling the truth
- Clinicians have the responsibility to take action when NAI is suspected

It is absolutely paramount to make a rapid diagnosis and treat the injuries, but most importantly to protect the patient from further harm, particularly vulnerable individuals (children and older adults). An early sign of abuse that is neglected may lead to further episodes and the potential of serious harm. For these reasons, procedures are usually in place and can be followed by passing on the problem to the appropriate team and professionals (see ***Chapter 44***).

Another important issue is the fact that any obvious injuries may provide important evidence regarding the mechanism, which may be important to a criminal investigation. The clinician must endeavour, without compromising treatment, not

to affect such evidence by their medical actions and bear in mind that forensic evidence may be needed for a conviction at a later stage. Furthermore, the importance of this is made more apparent if we consider that the victim of an attack may subsequently be a murder victim.

Patients

Apart from the deliberate circumstances outlined above, where the injury and mechanism are inconsistent, the clinician should consider the possibility that the patient may have an unknown pre-existing condition. For example, when a fracture occurs following what seems to have been an insufficient mechanism, the clinician should suspect that the bone was already susceptible to fracture. Such pathological fractures may be secondary to underlying problems such as metastatic or primary tumour, osteoporosis or congenital disease. One example is the preambulatory child with multiple fractures secondary to osteogenesis imperfecta, which may mimic NAI. Failure to identify a pathological fracture through a primary tumour will lead to inappropriate initial management compromising appropriate cancer treatment. An initial osteoporotic or herald fracture should signal the need for appropriate investigation and, potentially, treatment to prevent further or secondary fractures. Similarly, fractures may be secondary to an undiagnosed or poorly controlled medical condition. For example, a patient presenting with a scalp laceration and a wrist fracture may have fallen as a result of a transient ischaemic attack (a hidden patient factor). In this situation, it is essential to include a medical secondary survey to identify the real cause of the injuries sustained and prevent further trauma.

Injuries

When analysis of the formula 'mechanism + patient = injury' has failed to identify hidden injury, there are two other approaches:

1. the look everywhere approach;
2. the focused exclusion approach.

Look everywhere approach. This represents the secondary and tertiary elements of the ATLS system and involves a detailed secondary survey, from top to bottom and at different time points: soon after the initial treatment phase when measures relating to saving the patient's life have been completed, the day after injury, e.g. during a ward round, or several days after injury, e.g. when the patient first wakes up in the intensive care unit (ICU). The implementation of whole-body CT (WBCT) (scanning the whole body) in all major trauma centres has allowed the clinical team to pick up injuries early. Such injuries would have been missed in the past when reliance was placed on the initial radiographs of the chest, pelvis and cervical spine. The threshold for using more WBCT has been lowered substantially. There is no doubt that WBCT scan algorithms have been shown to accelerate diagnostic work-up, but their effect on survival is controversial. Moreover, concerns have been voiced about the overexposure of patients to radiation with the increasing and often uncritical use of this type of scan. The effective radiation dose to all organs from a single full-body CT is 12–16 millisieverts (mSv). Survivors of the atomic bomb whose radiation dose was in the range 5–100 mSv had a statistically significant increase in the risk of solid cancers. Overall, the risks associated with one scan are relatively modest, approximately 1 in 1250 or 0.08%. However, it has been reported that widespread liberal use of CT is responsible for 1.5–2.0% of all cancers in the USA. Of interest, WBCT equates to 76 chest radiographs or 6 months of background radiation. It has been suggested that it should be requested wisely and that developing a triaging protocol can minimise the criticism of its overuse.

Focused exclusion approach. This is based on the knowledge that some specific injuries are missed on a remarkably regular basis. Such injury patterns include metatarsal and metacarpal fractures, scaphoid fractures, perilunate dislocations and posterior shoulder dislocations. When such injuries are suspected, a detailed focused history, clinical examination and appropriate investigations should be carried out to either confirm or exclude them. A high level of alertness and a high index of suspicion are always required to think beyond the obvious.

Summary box 26.7

Trauma assessment

- Knowledge of timelines for important diagnoses is essential
- Initial assessment should focus on what kills first
- Screen high-risk patients before clinical signs become apparent, as it may be too late to intervene once signs develop

THE RESPONSE TO TRAUMA

Completion of the initial patient evaluation according to the formula (patient + mechanism = injury) should provide the necessary information to formulate and implement a 'response' (treatment). During this stage of management, the response to injury will continue to evolve and decompensation may occur unexpectedly. Vigilance is required throughout treatment to identify the potential exhaustion of biological reserve mechanisms.

The patient's response to injury

Immediately after the traumatic event, physiological reactions are initiated as part of the body's homeostasis mechanisms to preserve vital organ functions and to maintain survival. Initial responses represented by the so-called acute-phase response may be altered as the insult of injury evolves and deterioration of the patient's condition may occur.

It is essential therefore that the timing and nature of interventions should be altered accordingly. Reversal of haemodynamic instability due to ongoing bleeding must be carried out promptly in order to avoid the development of coagulopathy and secondary damage to vital organs (i.e. the brain) due to hypoxia. It is important to monitor the patient's physiological state, including body temperature, degree of oxygenation and organ perfusion. Low body temperature is commonly present

as a result of exposure, blood loss and dormancy. Covering the patient with appropriate blankets during transportation, resuscitation and surgery will reduce the risk of hypothermia, coagulation disturbances and ongoing bleeding. Administration of inspired oxygen or ventilation, if required, will improve the patient's degree of oxygenation.

Ongoing blood loss is associated with low blood pressure, reduced perfusion of the extremities (skin discoloration), tachycardia and an altered level of consciousness. Normally, vasoconstriction and endogenous clotting factors are activated to stop the bleeding in order to maintain adequate circulatory volume. A further consideration is that traumatised lung parenchyma cannot tolerate surplus fluid. Therefore, the latest resuscitation guidelines advocate a reduction in crystalloid administration and the early transfusion of blood products. Furthermore, there is a need to quickly identify the source of bleeding and stop loss of blood.

Another important part of the response to injury is activation of the immune–inflammatory system. Acute-phase mediators are released systemically, stimulating polymorphonuclear leukocytes to interact with the endothelium via the expression of surface receptors (integrins). If certain conditions are met extravasation of leukocytes may take place, particularly into the lung parenchyma, causing autodestruction. Clinical decisions should aim to minimise the risk of an exaggerated immune–inflammatory reaction. Surgical procedures, which can act as a second insult, where injury is considered the first, should be carefully timed and selected.

The medical response to injury

Initial management

After initial assessment of the patient's condition at the scene of an accident, paramedics communicate with the nearest hospital, triggering activation of 'the trauma team on call' and allowing personnel to expect the patient's arrival in the resuscitation room. The team leader, according to the ATLS protocol, will assign trained nurses and doctors to specific duties. Protective clothing, such as gloves and lead aprons, is required to protect the personnel from fluids and radiation exposure. Optimum coordination of the trauma team throughout the resuscitation process is essential to avoid careless delays, which may compromise the response time and the patient's condition. Timely involvement of different disciplines in assessing and planning treatment of injuries in different body areas is crucial and may lead to issues around priority in terms of planning and interventions. This situation could lead to confusion and uncertainty: 'Who should go first?' or 'What investigation should be next?' It is the role of the team leader to ensure that this is avoided and that decisions which may be critical for the patient's well-being are executed smoothly.

In situations where the system operates according to locally developed protocols, someone should have the responsibility of overruling the protocol if this is in the best interests of the patient, in order to keep the process moving along.

Following common pathways to manage patients can save time and reduce errors. In this respect the development of the NICE guidelines for the initial assessment and treatment of patients with major trauma can be useful. In addition, short standards documents, such as BOAST (British Orthopaedic Association Standards for Trauma), can be downloaded and printed for display in hospitals and so can support planning and decision making.

The development of local protocols, NICE guidelines and BOAST documents has contributed to the establishment of specific treatment pathways and to improved patient care and outcomes. Nonetheless, caution should prevail as, once a diagnosis (label) has been made, the pathway is set and it may not be in the patient's best interests if the diagnosis is misleading and troublesome. For instance, an older male patient fell in his garden and sustained a wrist fracture. He was given the label of 'accidental fall and wrist fracture', placed in a plaster of Paris back slab and arrangements were made for him to be admitted for fracture stabilisation. However, this patient had a number of medical comorbidities (heart failure, epilepsy, previous myocardial infarction and high blood pressure). Consequently, the label given as 'accidental fall and wrist fracture' may be associated with more severe underlying pathology and injuries. The wrong label may disguise the seriousness of the injuries sustained and the patient's condition may rapidly deteriorate, putting their life at risk. Thus, the first clinician in the diagnostic chain has disproportionate responsibility. Early inappropriate assessment and incorrect labelling could place the patient on the wrong treatment care pathway. This can be avoided by conducting appropriate physiological triage with senior input.

Beyond the first hour

The objective of the first (golden) hour is to reach the end points of resuscitation and completion of the diagnostic procedures identifying the injuries sustained. However, in polytraumatised patients, further interventions are necessary. For example, a spleen laceration, a lung contusion, a vertical shear pelvic fracture, an open right tibial fracture, a left femoral fracture or a distal humerus fracture are injuries necessitating treatment. The spleen laceration may be managed non-operatively or with embolisation, depending on its severity. The timing and priority of fixation of the other skeletal injuries and the type of fixation have been points of much discussion during the past decade.

Two treatment strategies have prevailed: early total care (ETC) and damage control orthopaedic (DCO) surgery. Since the late 1990s the ETC approach (fix all fractures early within 24 hours) revolutionised the management of patients with multiple injuries. This practice became the gold standard of treatment for polytrauma patients. Nonetheless, in specific groups of patients, for example those with severe chest and/or head injuries or those in an extreme physiological state (with ongoing bleeding from different sources such as the abdomen, pelvis and chest), it was observed that the ETC concept led to early complications and mortality.

With the advances made in the fields of molecular medicine, biology, intensive care and immunology the physiological response to injury was better understood. It was appreciated that the trauma sustained at the scene of an accident (first hit) induces an immune–inflammatory reaction that evolves with time and can prime the patient to reach an uncontrolled physiological state. Subsequently, surgical interventions can

induce an additional physiological stress (second hit) response, exhausting the biological reserve of the patient and leading to the development of adult respiratory distress syndrome (ARDS), multiple organ dysfunction syndrome (MODS) and mortality. These observations led to the acceptance of the so-called DCO philosophy, which is also called damage control surgery (DCS) in more generalised settings (see later chapters). The stages of DCO are:

- resuscitation;
- haemorrhage control;
- decompression;
- decontamination;
- fracture splintage.

Application of the DCO concept includes temporary stabilisation of long bone fractures and pelvis with external fixators. This approach allows rapid procedures to stabilise the affected bones as part of the resuscitation process. Definitive fixation of the fractures (conversion of the external fixators to intramedullary nailing for the femur and plating of the pelvis) would usually take place 4 days later, when the physiological state of the patient has been normalised, or even later if necessary. The two strategies of fracture fixation – the ETC and DCS – are currently practised on the basis of some specific criteria. The vast majority of polytrauma patients are suitable for ETC (80–90%). Specific criteria are shown in ***Table 26.2***.

TABLE 26.2 Criteria for damage control surgery (DCS) and early total care (ETC).

Criteria for DCS	Criteria for ETC
Hypothermia: <34°C	Stable haemodynamics
Acidosis: pH <7.2	No need for vasoactive/ inotropic stimulation
Serum lactate >5 mmol/L	No hypoxaemia, no hypercapnia
Coagulopathy	Serum lactate <2 mmol/L
Blood pressure <70 mmHg	Normal coagulation
Transfusion approaching 15 units	Normothermia
Injury severity score >36	Urinary output >1 mL/kg/h

Lately, other fixation strategies that have been proposed include the Early Appropriate Care, Safe Definitive Surgery and Prompt Individualised Safe Management (PRISM) concepts. The PRISM concept accepts that the decision-making process should be based on the principles of doing no 'further harm to the patient', intervening promptly and utilising the idea of individualised/personalised medicine. It is based on the understanding that every patient responds differently to the same degree of trauma, every individual has a different genetic profile and the fact that each trauma centre may have different resources and staff available to deal successfully with every trauma eventuality.

Irrespective of the treatment strategy selected, it is imperative to be aware that the patient's condition can change and deteriorate quickly. In particular, patients with a high ISS are at risk of decompensating rapidly. Therefore, ongoing evaluation of a patient's physiology is essential, with senior input involvement in order to deal successfully with the unexpected.

Overall, a plan should always be made available to the trauma team to inform them about the details of treatment, particularly in the operating theatre environment. The plan can be recorded on a whiteboard with clear guidance on the alternative pathway to be implemented should the patient's condition deteriorate. Ongoing monitoring of such parameters as core temperature, lactate, base excess and coagulation will provide knowledge of the patient's condition at any time point during care. Appropriate decisions can then be taken to ensure that the safety of the patient has not been compromised. For instance, if the ETC concept has been applied and a polytrauma patient with a head injury and chest trauma is on the operating table waiting to have a femoral fracture stabilised with intramedullary nailing and their condition deteriorates, the ETC plan can be changed to DCO (an external fixator can be applied quickly to stabilise the femoral fracture temporarily). This allows prompt transfer of the patient to the ICU, where resuscitation can continue within a safe environment with ongoing monitoring of vital organ function (lungs and brain).

When injuries involve the input of different disciplines in terms of surgical intervention, for instance a general surgeon for a liver laceration, a neurosurgeon for an intracranial haematoma, an orthopaedic surgeon for a pelvic and femoral fracture and a maxillofacial surgeon for a depressed orbital fracture, the most important intervention should go first. Clearly, communication and prioritisation among the team members is essential. As each procedure nears completion, communication with the anaesthetist will allow a decision as to whether it is safe to go on to the next procedure. Not infrequently, and assuming that it is technically possible, two teams can work simultaneously to significantly shorten the time the patient will spend in the operating theatre and allow subsequent prompt transfer to the ICU for optimisation and ongoing monitoring.

LOCAL PROTOCOLS AND GUIDELINES

While the ATLS protocol has become the standard of care for the initial management of patients with multiple injuries, other protocols and guidelines have also been developed to facilitate the treatment of patients in a more standardised way. Nowadays it is common practice for trauma centres to develop their own local protocols and guidelines, although national guidelines may also exist. However, regional developed guidelines may refer to smaller areas of clinical practice, such as antibiotic prophylaxis for open fractures, mass transfusion, pharmacotherapy for coagulation disturbances, steroids for spinal cord injuries, clearance of the cervical spine and angiographic embolisation of pelvic fractures or abdominal injuries. These protocols can facilitate swifter decision making, eliminating delays and benefiting the patient. They also protect the clinician and care provider with regard to medicolegal issues. An example relating to angiographic embolisation of pelvic fractures is shown in ***Figure 26.7***.

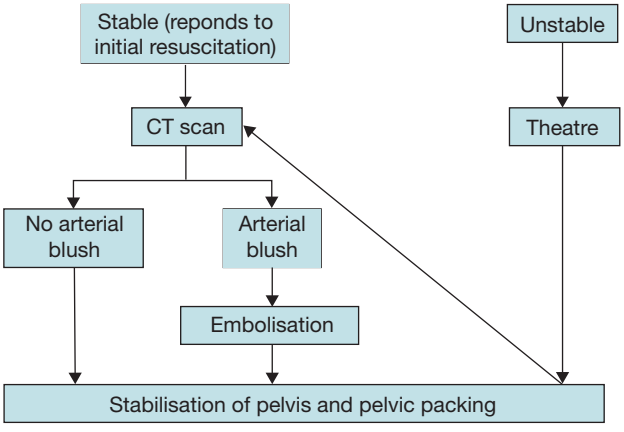

Figure 26.7 Pelvic fracture: angiography protocol. CT, computed tomography.

Local policies focusing on the creation of single charts facilitating daily input of a patient's vital signs and biochemical results are also useful in allowing sequential observation of the results, which can demonstrate important trends. These trends can be useful in identifying, at an early stage, a clinical condition that can be treated within the timeline concept and prior to irreversible damage to the affected organ, at which point any form of intervention will be meaningless. For instance, the clinical evolution of respiratory insufficiency in an individual without pre-existing lung disease secondary to pulmonary embolism is easier to identify by evaluating the trend in the oxygen saturation of inspired oxygen.

Planning an individual operation

Operative procedures in a multiply injured patient in a critical physiological state can be essential for their survival. It is vital to make the right decision the first time. This can be successfully done by studying in detail the history, the patient's condition and any interventions made, followed by appropriate planning and execution of treatment. For example, analysing the biochemical markers, overall physiological state and fracture pattern and deciding factors such as patient positioning, the surgical approach to be used to gain adequate access to a bone for reduction, the type of implant to be used, the correct positioning of the implant, soft-tissue handling and the type of rehabilitation are just some of the parameters for consideration.

Summary box 26.8

The response to trauma

- Rationalise patient management with the development of protocols and guidelines
- Avoid unnecessary delays
- Observe trends and promptly identify evolving conditions

The surgeon should ask themselves (and ensure that they have answered) questions such as: Is it the optimum time to carry out the procedure? Am I the most appropriate person to carry out the procedure? Have I made a contingency plan should the procedure become difficult to complete? What is my plan A, B and C so I am able to successfully finish the procedure? Recording all the important issues on a piece of paper, or suitable alternative, will ensure that no parameter of importance is left out of the planning. In order that the procedure runs smoothly in theatre, the plan can be documented on the whiteboard. Such a practice will allow every member of the surgical team to be aware of the potential issues to be addressed and the plan of action that has been decided. The appropriate surgical equipment for plans A, B and C can be clearly identified and kept in close proximity to the operating theatre. The above strategy will eliminate unnecessary detail and ensure that there will be no surprises, for instance essential equipment being unavailable.

The response to the mechanism of injury (injury prevention)

Not infrequently, two or more patients are seen who have been injured in the same geographical area by the same mechanism. A possible scenario is that, over a period of a few weeks, several patients are noted to present with a fall of 2–3 metres in the same location, for example from a particular bridge. Further investigation might establish that the bridge is easy to access, perhaps because of damage to safety measures. With appropriate steps the damage can be rectified and access to the bridge limited, preventing further harm. In this example, the surgeon is involved in both the treatment and the prevention of injury. Therefore, when a mechanism of injury becomes more frequent and is associated with serious life-threatening trauma, it is essential to take steps either to eliminate the mechanism or to lessen the consequences.

Summary box 26.9

Planning an individual operation

- Adequate preoperative planning is essential to eliminate unnecessary delays
- Document the plan to ensure no important parameter is left out
- A whiteboard can be used to demonstrate the order of plan execution and to act as a means of communication among staff

The issue of injury prevention is an important consideration not only for clinicians but also for other stakeholders, for example politicians (developing appropriate legislation relevant to road safety), manufacturers (optimising the safety of cars) and the construction industry. This accounts for the improvements in safety that can be achieved.

The response to patient factors

Injuries presenting with increased frequency in an individual patient require special attention. Older patients, perhaps with multiple medical problems, can represent such a vulnerable group. The reduced bone mineral density making their

Summary box 26.10

Response to the mechanism of injury

- Be alert to identifying frequent patterns of injury
- Many stakeholders, including clinicians, should engage in injury prevention
- Legislation and education are essential

skeletons 'fragile' is associated with an increased risk of fracture. Hip fractures, wrist fractures, spinal fractures and pelvic fractures are just some of the many results following minimal trauma. The risk of fracture is increased with the presence of comorbidities such as visual impairment, Parkinsonism or transient ischaemic heart attacks. Consequently, knowing the parameters associated with a particular group of patients, attempts can be made to reduce the number of patients requiring treatment. For instance, prescription of bone protection therapy and addressing medical comorbidities early (e.g. cataract surgery) may help to reduce the number and severity of injuries sustained. Development of specific guidelines and protocols can facilitate the much-needed implementation of unified measures allowing easy patient access and treatment.

Summary box 26.11

Response to patient factors

- Older patients may be vulnerable to injury, owing to poor bone stock and other comorbidities
- Bone protection treatment is essential
- Development of local and national guidelines to respond to patient factors is desirable

CONCLUSION

Trauma can affect all patient age groups. The severity of injury depends on the type and nature of the mechanical force applied. The formula 'patient + mechanism = injury' should be kept in mind when dealing with trauma patients. There is a connection among the three components of the formula, all of which should fit together in a coherent way. When no relationship can be made, examine for pre-existing pathology and hidden injuries or consider whether the history given by the patient is at fault (including deliberate attempts to mislead).

Be aware of the timeline concept because there is a minimum response time to initiate and complete treatment in order to deal with a specific condition successfully.

In patients with multiple injuries early assessment and management is performed using the ATLS protocol and other guidelines developed either locally or at a national level. In the early clinical pathway of the patient the objective is to restore haemodynamic stability (save life – prevent uncontrollable bleeding and death) and minimise the risk of developing subsequent complications. Surgical interventions should be short and part of the resuscitation process, executed in a timely manner. Afterwards, treatment for definitive stabilisation can be personalised for optimum outcomes. Existing specific criteria can aid the clinician to make the right decision for the right patient at the right time. Different groups of patients, such as older patients and children, have different demands and should be managed accordingly. While clinicians are focused on treating the injury sustained, they have other responsibilities, including active involvement in preventive measures. Preventive measures should be commissioned when particular mechanisms can be identified as being common or important causes of injury.

Summary box 26.12

Conclusion

- Look for hidden injuries
- Remember the timeline concept
- Different treatment strategies exist, complementing each other
- Specific criteria will decide the fixation strategy: ETC or DCO

FURTHER READING

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